

AI Social Media Manager

Complete Product Paper · Bootstrap Release

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What this document covers: Complete feature specification, AI agent architecture, multi-tenant data model, integration catalogue, dashboard overview, pricing model, deployment requirements, and product roadmap for the AI Social Media Manager platform.

1. Executive Summary

AI Social Media Manager is a multi-tenant SaaS platform that fully automates the marketing operations of a brand's digital presence. It replaces a marketing team's routine execution work — content creation, scheduling, customer service responses, email campaigns, and performance analysis — with a coordinated system of AI agents, each specialising in a single discipline.

The platform is not a content suggestion tool. It acts. It generates content, reviews it against the brand's own guidelines, schedules it, monitors engagement, handles customer inquiries, assembles weekly newsletters, optimises product listings, and flags the things that genuinely need a human decision. The human manager sees a dashboard of outcomes, not a queue of tasks.

The system is designed for agencies managing multiple brands and for individual businesses that want enterprise-grade marketing automation without an enterprise-sized team.

2. The Problem It Solves

Running consistent, high-quality digital marketing for a brand requires:

- Daily content creation across 2–5 platforms
- Brand voice consistency across every post, email, and customer reply
- Real-time response to customer inquiries (expected within hours)
- Weekly email newsletters and campaign planning
- Monthly performance analysis and strategy adjustment
- Constant A/B testing of subject lines, posting times, and content angles

A single marketing manager can do this for one brand, imperfectly, while burning out. An agency doing this for 10 clients needs 10 people. Neither scales.

AI Social Media Manager reduces this to: **review what needs a human, approve it or override it, and let the system handle the rest.**

3. Core Architecture

The platform is built on a multi-agent orchestration model. Seven specialised AI agents run concurrently, communicating through a priority-based job queue (BullMQ on Redis). Each agent has a single responsibility and cannot act outside its scope.

Layer	Technology	Purpose
API Server	Node.js / Express	HTTP endpoints, webhook handling, auth

Job Queue	BullMQ + Redis (Upstash)	Async task routing with priority levels
AI Engine	Anthropic Claude API	Sonnet 4.6 (complex) + Haiku 4.5 (fast)
Relational DB	Supabase (PostgreSQL)	Metrics, schedules, escalations, tenants
Document DB	MongoDB Atlas	Content, customers, decisions
Dashboard	Next.js 15 (Vercel)	Management UI with Supabase Auth

Prompt caching is implemented throughout — cached system prompts (brand guidelines, skill instructions) reduce Anthropic API costs by 80–90% on repeated calls within the same tenant context.

Job Priority Levels

Priority	Level	Triggered by
URGENT (1)	Highest	Angry customer detected, API failure, brand crisis
HIGH (5)	High	Customer service, live engagement response
NORMAL (10)	Standard	Content generation, email campaigns
LOW (20)	Background	Analytics aggregation, reporting

4. The Seven AI Agents

4.1 Content Generator

Generates all written content for the brand, on schedule and on demand.

Output types:

- Instagram and Facebook captions — 5 variations per brief, ranked by quality with engagement hooks
- Email campaign copy — complete body with 2–3 subject line A/B variants and preview text
- Blog posts — SEO-optimised, 500–700 words, HTML-ready with meta description and slug
- Product descriptions — benefit-focused, luxury-copy style, 150 words max
- 30-day content calendars — platform-by-platform breakdowns with key dates and email recommendations

How brand context works: Every piece of content is generated using the tenant's brand guidelines as a cached system prompt. The model has no hardcoded assumptions — a fashion brand in Lagos and a fintech in London get entirely different output from the same system. After generation, the best variation is automatically routed to the Brand Guardian for review.

4.2 Brand Guardian

The quality gate. Reviews every piece of content before it leaves the system. Operates in two stages.

Stage 1 — Static checks (instant, zero cost):

- Character limit validation per platform (Instagram: 2,200 · Facebook: 63,206 · Twitter: 280)
- Banned phrase detection (configured per tenant)
- FTC compliance flags — #ad / #sponsored disclosure detection
- All-caps, excessive punctuation, and prohibited claims detection

Stage 2 — Claude deep review:

- Brand voice consistency score (0–100)
- Tone analysis against the tenant's personality traits
- Specific issue identification with actionable written feedback
- Revised content suggestion when score is below threshold

Decision outputs:

Decision	Meaning	Next action
approved	Strong, on-brand	Routed to Social Media Manager for scheduling
approved_with_suggestions	Good, minor polish noted	Published, suggestions logged
needs_revision	Claude provides corrected version	Re-queued for re-review
rejected	Cannot be published	Escalated to human manager with reason

4.3 Social Media Manager

Handles everything that happens after content is approved.

- Post scheduling via Buffer — Instagram, Facebook, Twitter, TikTok, Pinterest
- Optimal posting time calculation — analyses historical engagement, recommends best slots per platform
- Cross-platform adaptation — reformats one approved piece for each platform's character limits and best practices
- Hashtag strategy — rotates sets, avoids repetition, tracks performance per tag
- Engagement monitoring — detects high-engagement posts and surfaces them for amplification
- Sentiment analysis on incoming mentions and comments

4.4 Email Strategist

Full email marketing automation from brief to send.

- Campaign creation from a brief (goal, segment, offer) to complete ready-to-send email
- Subject line A/B testing — 2–3 variants: curiosity, FOMO, benefit-driven, exclusivity angles
- Subscriber segmentation — new, engaged, repeat, VIP, at-risk, inactive
- Send time optimisation per segment based on historical open rates

- Weekly newsletter assembly every Sunday — curated from week's performance and product updates
- Campaign performance analysis — open rate, click rate, revenue attribution

4.5 Customer Service Agent

Handles incoming customer inquiries across all connected channels.

Channels:

- Instagram DM, WhatsApp, Website chat (Tidio), Email

Capabilities:

- Intent classification — purchase inquiry, complaint, shipping question, product info, return request
- Sentiment detection — routes distressed customers to URGENT priority
- Brand-voiced responses — replies match the tenant's tone exactly
- FAQ integration — checks knowledge base before composing a response
- Escalation triggers — sentiment below threshold, VIP customers, order disputes

Scope boundary: The agent does not make refund decisions, does not promise delivery dates, and does not handle complaints requiring account access. Those escalate with full context attached.

4.6 Analytics Monitor

Aggregates performance data from all connected channels and surfaces actionable insights. Runs at 18:00 daily per tenant timezone.

Output	Description
Daily metrics rollup	Reach, engagement rate, conversions, revenue — stored in Supabase
Anomaly detection	7-day rolling average; alerts when any metric moves $\pm 25\%$
30-day forecasts	Predicted performance based on trend analysis
Optimal posting times	Recalculated weekly from engagement data
Sales spike correlation	Traces revenue spikes to the content or campaign that drove them
Weekly performance report	Plain-English summary: what worked, what didn't, what to do next

4.7 E-commerce Optimizer

Improves product performance on the tenant's e-commerce platform. Supported: Shopify (full API), WooCommerce, BigCommerce, Wix (adapters ready).

- Product listing optimisation — rewrites titles, descriptions, and bullets for SEO and conversion
- Conversion funnel analysis — identifies drop-off points, suggests targeted fixes
- Demand forecasting — 90-day inventory predictions with stockout alerts
- Product recommendation engine — cross-sell and upsell suggestions from purchase history

- Pricing analysis — competitor-aware intelligence within configured guardrails

5. Multi-Tenant Architecture

The platform is built for multiple independent brands from the ground up. Every record in every database is scoped to a tenant. No tenant can see another tenant's data — this is enforced at the database layer, not just the application layer.

Isolation Layers

Layer	Mechanism
Supabase / PostgreSQL	Row-Level Security on every table · tenant_id column · JWT app_metadata claim
MongoDB	tenantId field on all documents · all queries include { tenantId } filter
Redis / BullMQ	tenantId embedded in every job payload · skills extract before processing
Brand Config	Per-tenant JSONB in brand_configs · 5-min in-memory cache per tenant
Credentials	Per-tenant rows in tenant_credentials · never in environment variables
Dashboard	Supabase JWT with tenant_id in app_metadata · middleware enforces routing

Authentication

- **Dashboard users:** Supabase JWT with app_metadata.tenant_id claim — issued on login, verified on every API request
- **Direct API access:** API_SECRET_KEY env var → DEFAULT_TENANT_ID (backwards-compatible for single-tenant setups and testing)

Brand Config Versioning

Every change to a tenant's brand guidelines creates a new version record in brand_configs with is_active=true and all previous versions deactivated. This provides a full audit trail and one-step rollback capability. Claude's prompt cache automatically invalidates when the guidelines change (different content = different cache key).

6. Dashboard

The management interface for human operators. Built in Next.js 15 (App Router) with Supabase Auth. Deployed to Vercel. Communicates with the Express API via Supabase JWT.

Screen	Purpose
Login / Signup	Email and password authentication via Supabase. Email confirmation on signup.
Onboarding Wizard	5-step brand setup: company → brand voice → audience → integrations → launch

Overview	Live stats: open escalations, recent task activity, scheduled posts count
Content	Scheduled content calendar with platform, type, date, and status per post
Escalations	Full list of open and resolved escalations with context and human notes
Brand Settings	Live editor for brand config — identity, voice, audience. Saves to API.
Integrations	Connect Buffer, Mailchimp, Shopify/WooCommerce/BigCommerce/Wix, Meta, GA4

Onboarding Wizard — New Tenant Flow

Step	Fields collected
1 · Company	Brand name, tagline, industry, primary market, currency, website URL
2 · Brand Voice	Personality traits (multi-select), tone description, do list, don't list
3 · Audience	Primary audience description, secondary audience description
4 · Integrations	Platform connections (can be skipped and completed later in Settings)
5 · Launch	Review summary, activate tenant — automated operations begin immediately

7. Integration Catalogue

Service	Category	Status	Function
Buffer	Social	Live	Posts to Instagram, Facebook, Twitter, TikTok, Pinterest
Meta API	Social Data	Live	Instagram DMS, mentions, page analytics
Mailchimp	Email	Live	Campaign creation, list management, analytics
Shopify	E-commerce	Live	Products, orders, inventory, analytics
WooCommerce	E-commerce	Stub	Adapter interface ready, credentials accepted
BigCommerce	E-commerce	Stub	Adapter interface ready, credentials accepted
Wix	E-commerce	Stub	Adapter interface ready, credentials accepted
Google Analytics 4	Analytics	Stub	Property ID accepted, aggregation pending
Tidio	Customer Service	Stub	Webhook handler live, response sending pending
Supabase	Database	Live	All structured data — metrics, schedules, escalations
MongoDB Atlas	Database	Live	Content history, customer profiles, decisions
Upstash Redis	Queue	Live	BullMQ job queues with priority routing
Anthropic	AI	Live	Claude Sonnet 4.6 + Haiku 4.5 for all AI tasks

Stub status: Credential storage is live (tenants connect via the dashboard and credentials are stored encrypted in Supabase). The adapter interface exists with the correct method signatures. The actual API call implementations are the next development step — no architectural changes required.

8. Automated Schedule

All times in the tenant's configured timezone (default: Africa/Lagos, WAT UTC+1). The scheduler queries all active tenants on each trigger and enqueues a separate job for each — every tenant gets their own isolated execution.

Time	Task	Output
08:00 daily	Generate daily content	Instagram + Facebook captions → Brand Guardian → Buffer
18:00 daily	Aggregate metrics	All channels → anomaly detection → daily_metrics table
18:00 Sunday	Weekly newsletter	Assembled newsletter → A/B subject lines → Monday send
00:00 daily	Health check	System status logged
On demand	Customer inquiry	Webhook-triggered, response within minutes
On demand	Manual trigger	Any job via dashboard or direct API call

9. Escalation System

The system knows what it can and cannot handle. Anything requiring human judgment is escalated — not silently dropped. Every escalation includes full context so the human manager can act immediately.

Escalation Triggers

- Brand Guardian quality score below 50 (high-risk content)
- Content flagged as requiring mandatory human approval (sensitive topics, large campaigns)
- Customer sentiment below threshold — angry, distressed, or threatening language detected
- API failures on critical integrations (Buffer, Mailchimp) with retry exhausted
- Anomaly detection alert — metric moves beyond $\pm 25\%$ of 7-day rolling average
- Customer requests outside AI scope — refunds, account access, complex disputes

Escalation Data

Every escalation record includes: type, reason, skill that raised it, original job ID, truncated content or context (first 200 characters), and timestamp. Resolved escalations show the human manager's notes.

Escalation Delivery

- **Dashboard** — Escalations page shows open items in real time (Supabase live query)

- **Webhook** — Configurable URL (Slack, WhatsApp, email) notified immediately on escalation via `ESCALATION_WEBHOOK_URL`

10. Data Model

Supabase (PostgreSQL) — Structured & Relational

Table	Purpose
tenants	Tenant registry — name, slug, plan, status
brand_configs	Versioned brand guidelines per tenant (JSONB) — identity, voice, audience, compliance
tenant_credentials	API keys per service per tenant — stored as JSONB, cached 10 min
platform_connections	Connection status per platform per tenant with metadata
onboarding_progress	Wizard step completion state per tenant
tenant_api_keys	Hashed API keys for direct API access (alternative to Supabase JWT)
task_log	Every job that ran — skill, action, status, duration, tenant_id
escalations	Human review queue — type, reason, payload, resolved status, human notes
content_schedule	Scheduled posts — platform, type, scheduled_at, status, mongo reference
email_campaigns	Campaign records — Mailchimp ID, subject, segment, open/click rates, revenue
faq	Customer service knowledge base per tenant — question, answer, hit count
daily_metrics	Time-series performance — date, channel, metric_key, value per tenant

MongoDB (Document) — Flexible Schema

Collection	Purpose
Content	Generated content with all variations, brand review results, performance metrics, jobld
Customer	Customer profiles — inquiry history, purchase history, segmentation, LTV, NPS
Decision	Full AI decision log — inputs, outputs, skill, duration, escalation flag
Metrics	Channel performance snapshots with mixed schema per channel type

11. Pricing Model

Based on per-tenant Claude API cost analysis. The AI processing cost per tenant runs approximately \$2–15 per month depending on usage intensity, enabling strong margins at modest subscription prices.

Tier	Price/month	Included	Claude cost es
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Starter	\$79	1 brand · 3 platforms · 2 email campaigns/month	~\$2–5
Growth	\$149	1 brand · all platforms · unlimited campaigns · analytics reports	\$5–15
Agency	\$499	Up to 10 brands · all features · priority support	~\$20–100
Enterprise	Custom	Unlimited brands · SLA · dedicated onboarding	Variable

Unit economics at 50 tenants (Growth tier):
Revenue: \$7,450/month · Infrastructure: ~\$450/month · Claude API: ~\$375/month ·
Gross margin: ~89%

12. Deployment

Production Stack (Recommended)

Component	Service	Cost/month
API + Workers	Railway (1 service)	\$5–12
Dashboard	Vercel (Hobby/Pro)	\$0–20
Redis	Upstash (free → paid)	\$0–15
PostgreSQL + Auth	Supabase (free → Pro)	\$0–25
MongoDB	Atlas M0 → Flex	\$0–20
AI Processing	Anthropic API	Per usage (~\$2–15/tenant)
Total (bootstrap)	Pre-revenue	~\$5–72 fixed

Local Development

Two processes — no Docker required:

```
# Terminal 1 – API server (port 3000) npm run dev # Terminal 2 – Dashboard (port 3001) cd dashboard && npm install && npm run dev
```

The only cloud dependency during local development is Supabase — the API connects to your Supabase project for auth and data. Redis (Upstash free tier) and MongoDB Atlas (M0 free) are also cloud-based.

Required Environment Variables

Variable	Required	Purpose
ANTHROPIC_API_KEY	Yes	Claude API access — all AI capabilities
SUPABASE_URL	Yes	Supabase project URL

SUPABASE_SERVICE_KEY	Yes	Supabase service role key — bypasses RLS
MONGODB_URI	Yes	MongoDB Atlas connection string
REDIS_URL	Yes	Upstash Redis URL (rediss:// format)
API_SECRET_KEY	Recommended	Bearer token for direct API access
DEFAULT_TENANT_ID	Recommended	Tenant UUID for API_SECRET_KEY auth
ESCALATION_WEBHOOK_URL	Optional	Slack/WhatsApp/email for escalation alerts
ALLOWED_ORIGINS	Production	Comma-separated CORS origins (dashboard URL)
NEXT_PUBLIC_SUPABASE_URL	Dashboard	Supabase URL for Next.js client
NEXT_PUBLIC_SUPABASE_ANON_KEY	Dashboard	Supabase anon key for auth
NEXT_PUBLIC_API_URL	Dashboard	Express API base URL

13. What the System Does Not Do (v1.0)

Being explicit about scope prevents misunderstandings and sets correct expectations for both operators and tenants.

- **No financial decisions** — no refunds, no pricing changes, no budget allocation without human approval
- **No posting without connected integration** — content is queued; if Buffer is not connected, it waits
- **No model training on your data** — all Claude calls are stateless; no fine-tuning occurs
- **No voice or video content** — text and static image captions only in v1.0
- **No paid advertising** — organic content only; no Meta Ads or Google Ads integration
- **WooCommerce, BigCommerce, Wix** — credential storage live, API operations are stubs
- **GA4** — property ID is stored and accepted; data aggregation is a stub
- **Tidio** — webhook receiver is live; outbound response sending is a stub

14. Product Roadmap

Phase	Focus	Key deliverables
Phase 7	Complete stubs	WooCommerce, BigCommerce, Wix adapters · GA4 aggregation · Tidio sending
Phase 8	Billing	Stripe integration · per-tenant subscriptions · usage metering · plan enforcement
Phase 9	Advanced analytics	AI-generated reports · competitor benchmarking · content performance prediction
Phase 10	Content expansion	TikTok/Reels scripts · image briefs · multilingual (French, Swahili, Yoruba, Arabic)
Phase 11	Agency tools	Sub-account hierarchy · white-label dashboard · client PDF exports · approval workflow

Phase 12	Enterprise	SSO · audit logs · dedicated infrastructure · SLA · custom model fine-tuning
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15. Key Design Principles

The AI handles execution. Humans handle judgment.

No content is published that hasn't passed automated brand review. No customer situation is resolved if it requires discretion. The system escalates generously rather than guessing.

Tenant isolation is absolute.

A bug affecting one tenant cannot expose another tenant's data. Every query, job, and cached value is scoped. Isolation is enforced at the database layer (RLS + field filtering), not just the application layer.

Graceful degradation.

If Supabase is unavailable, the system falls back to default brand config and logs a warning. If MongoDB is unavailable, content is generated but not persisted — the job completes without crashing. No single service failure takes down the whole system.

Cost efficiency by design.

Prompt caching on every Claude call. Brand guidelines (the largest cached block) are reused across all calls for the same tenant. Heavy reasoning uses Sonnet; quick classification uses Haiku. Profitable at low per-tenant pricing.

Everything is auditable.

Every AI decision is logged to MongoDB with inputs, outputs, skill, job ID, and duration. Every escalation is logged with context. Every credential change is timestamped. The human manager can always see what the AI did and why.

This document reflects the current state of the codebase as of Phase 6 (bootstrap release). The system is fully operational for single and multi-tenant deployments on Railway + Vercel with Supabase and MongoDB Atlas.